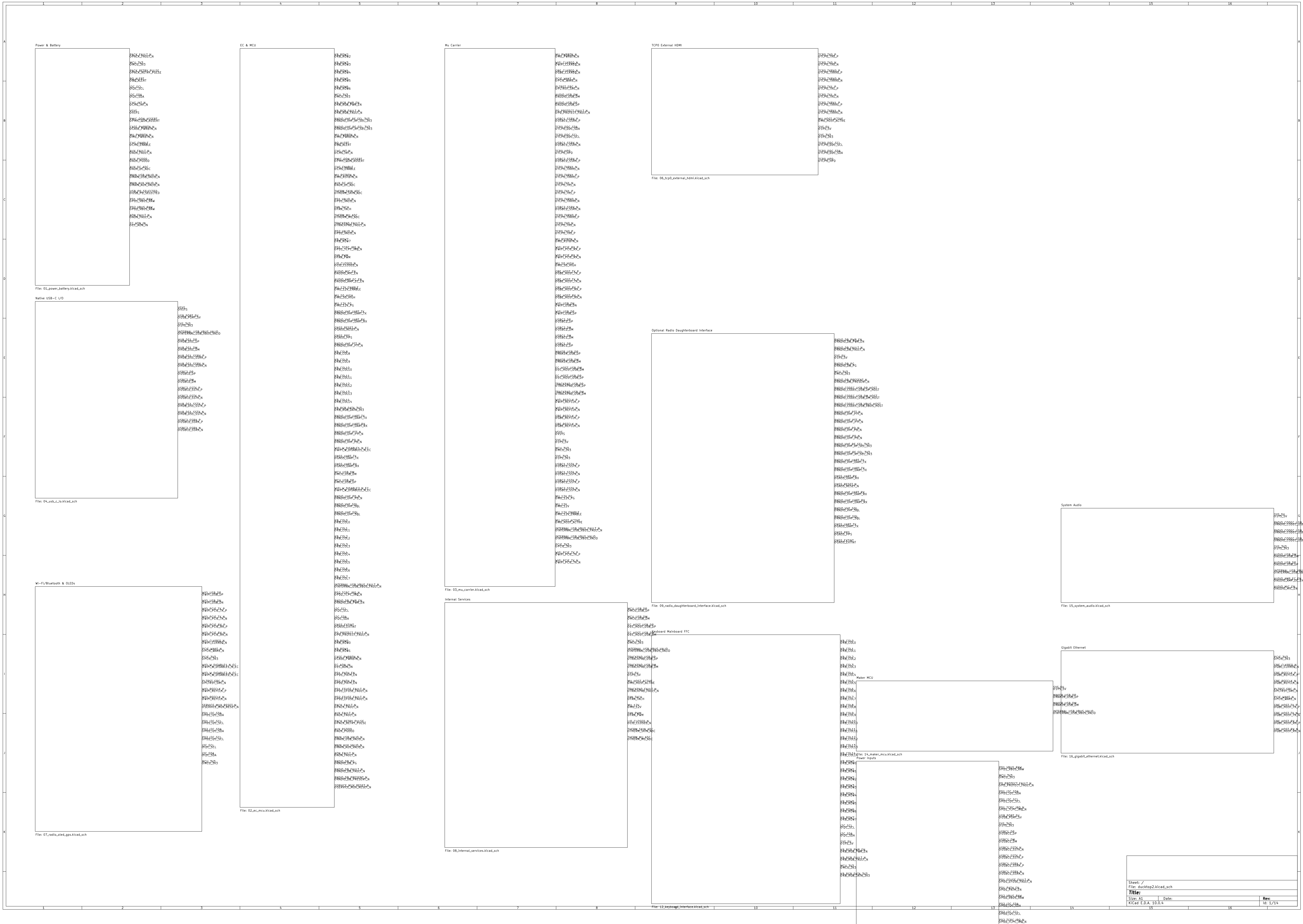
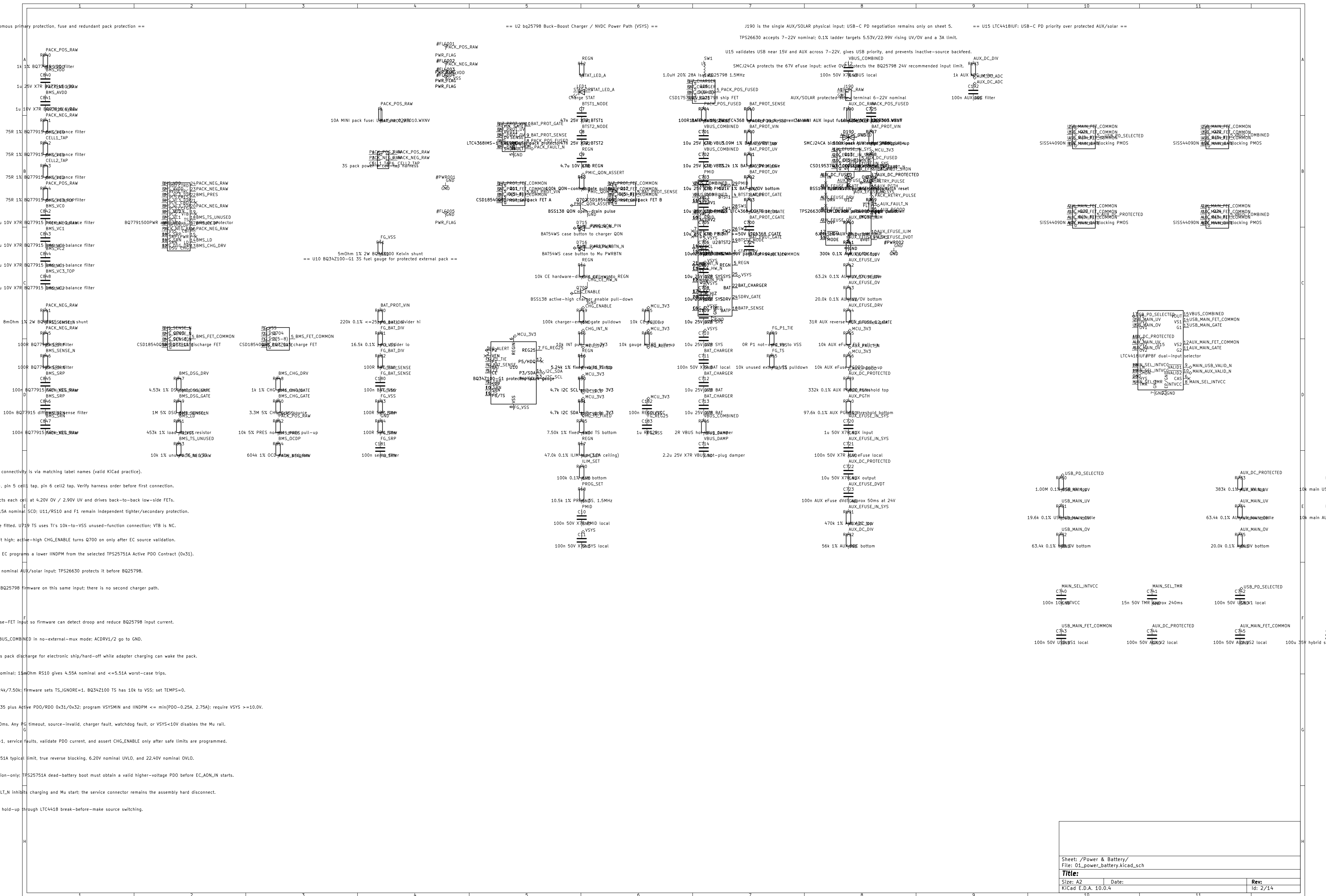






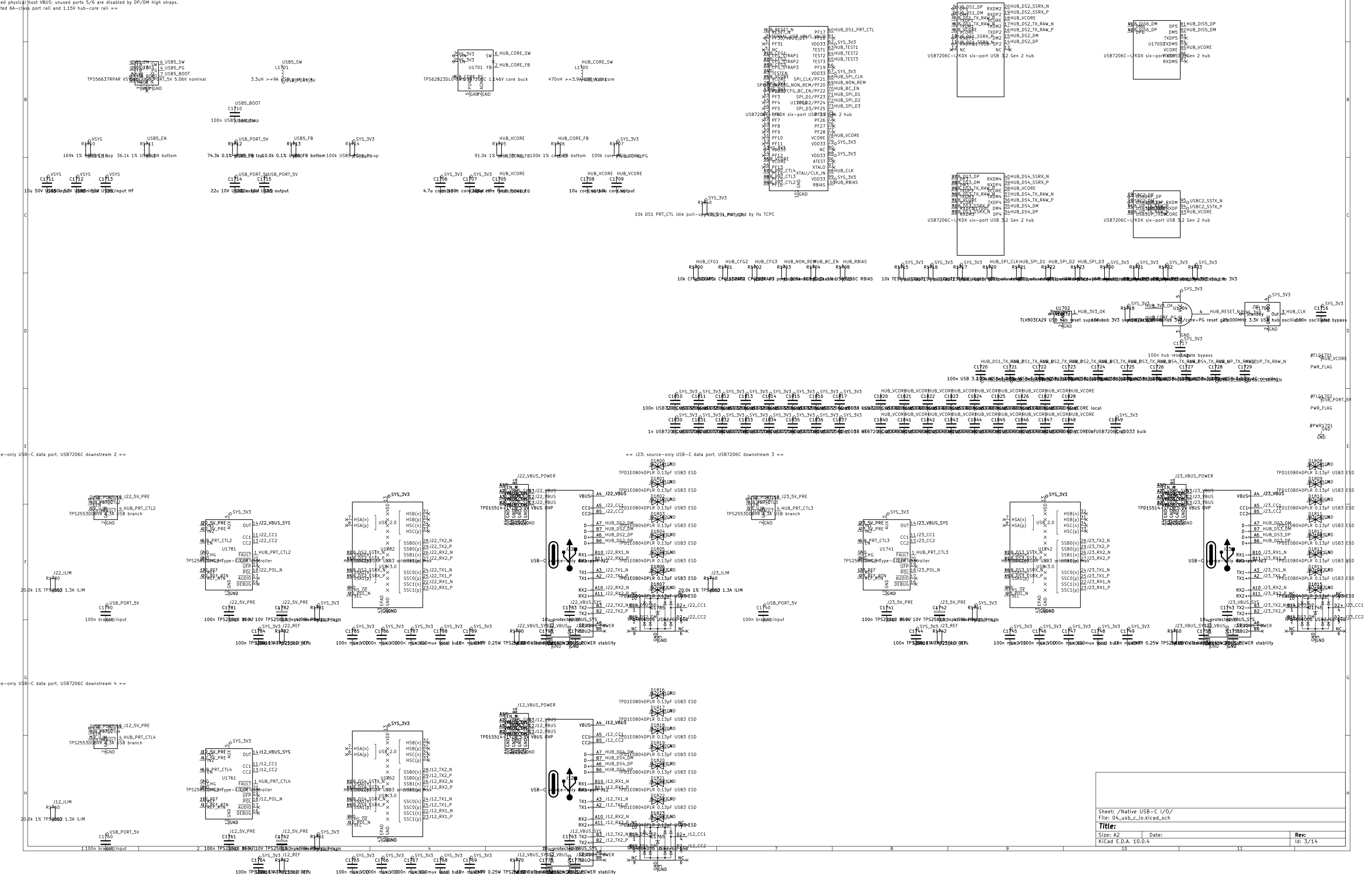
architectures: USB7206C hub and three source-only Type-C ports ==

the hub upstream. Hub port 1 feeds rear-right dual-role J11 on sheet 5.

1S514-1 blocks forced high-voltage VBUS; compliant charger attach is a safe no-op.

ports; per-port current limiting prevents one cable fault from dropping the laptop.

ed physical host VBUS; unused ports 5/6 are disabled by DP/DM high straps
ted 6A-class port rail and 1.15V hub-core rail ==

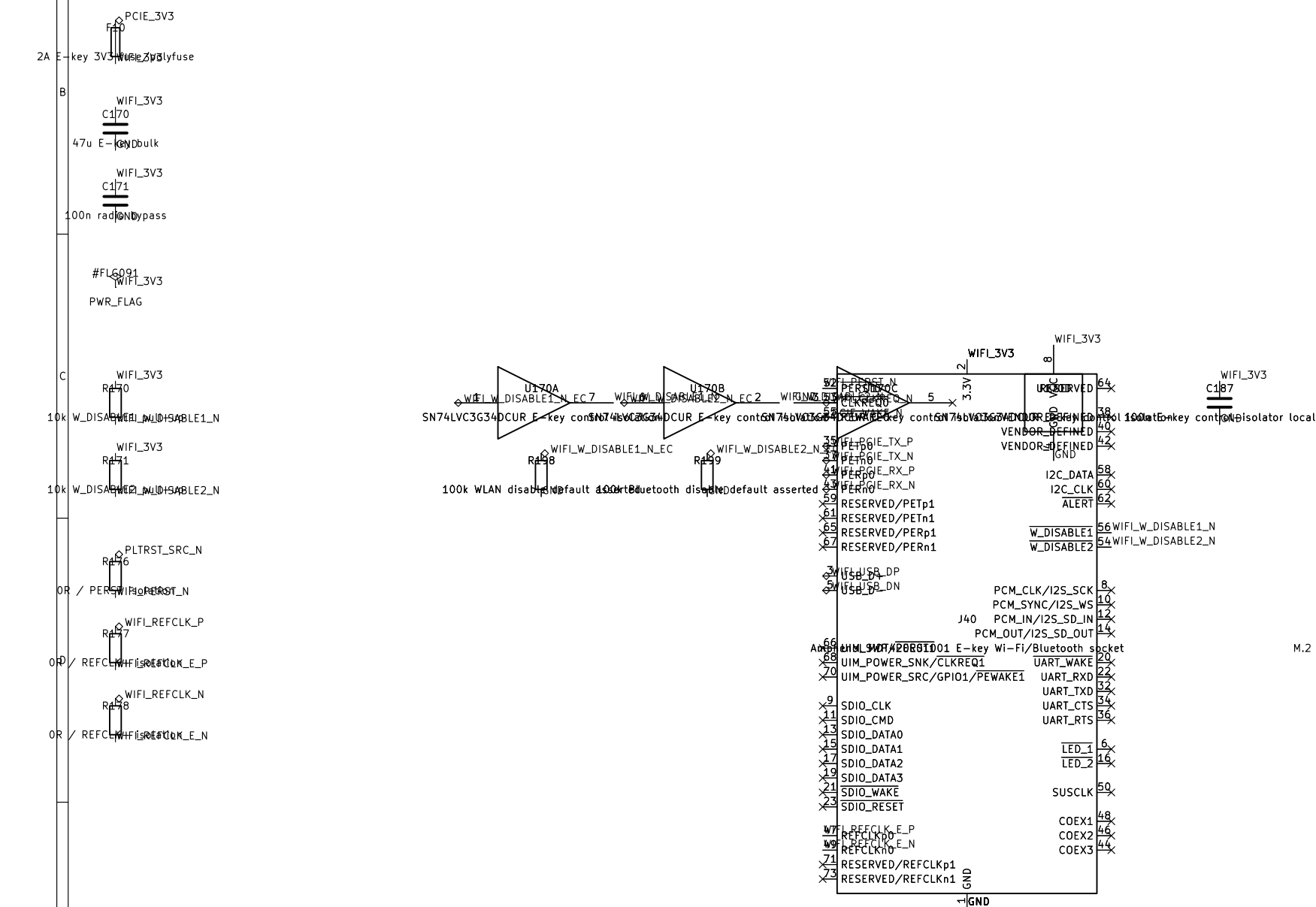


Wi-Fi/Bluetooth and dual OLED peripherals ==

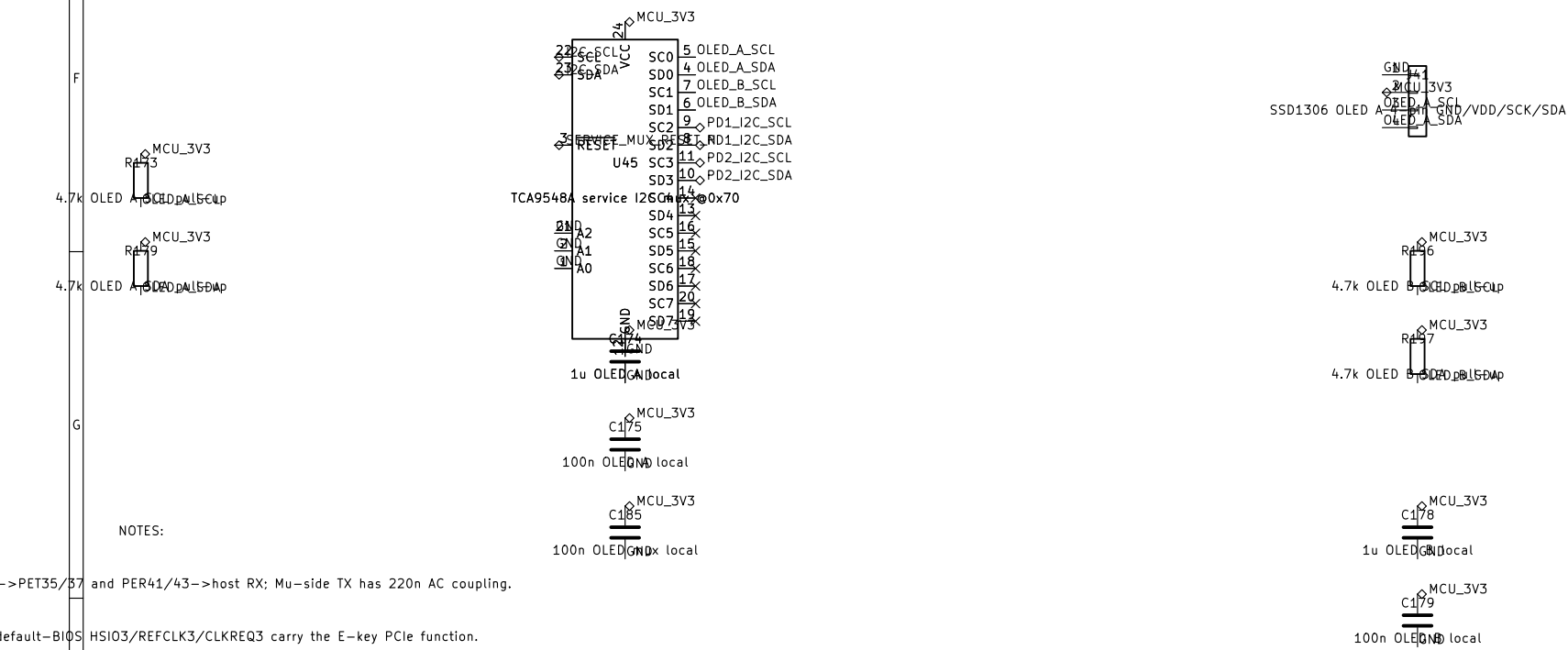
GNSS and amateur-radio hardware moved to the optional radio daughterboard.

the always-on EC service bus and work without the radio daughterboard.

J40 M.2 E-key Wi-Fi/Bluetooth module ==



```
al SSD1306-2C OLED/status display headers ==
```



NOTES:

->PET35/37 and PER41/43->host RX; Mu-side TX has 220n AC coupling.

2 2230 Key-E: Wi-Fi over PCIe and Bluetooth over USB; AX211/CNVio2 is prohibited.

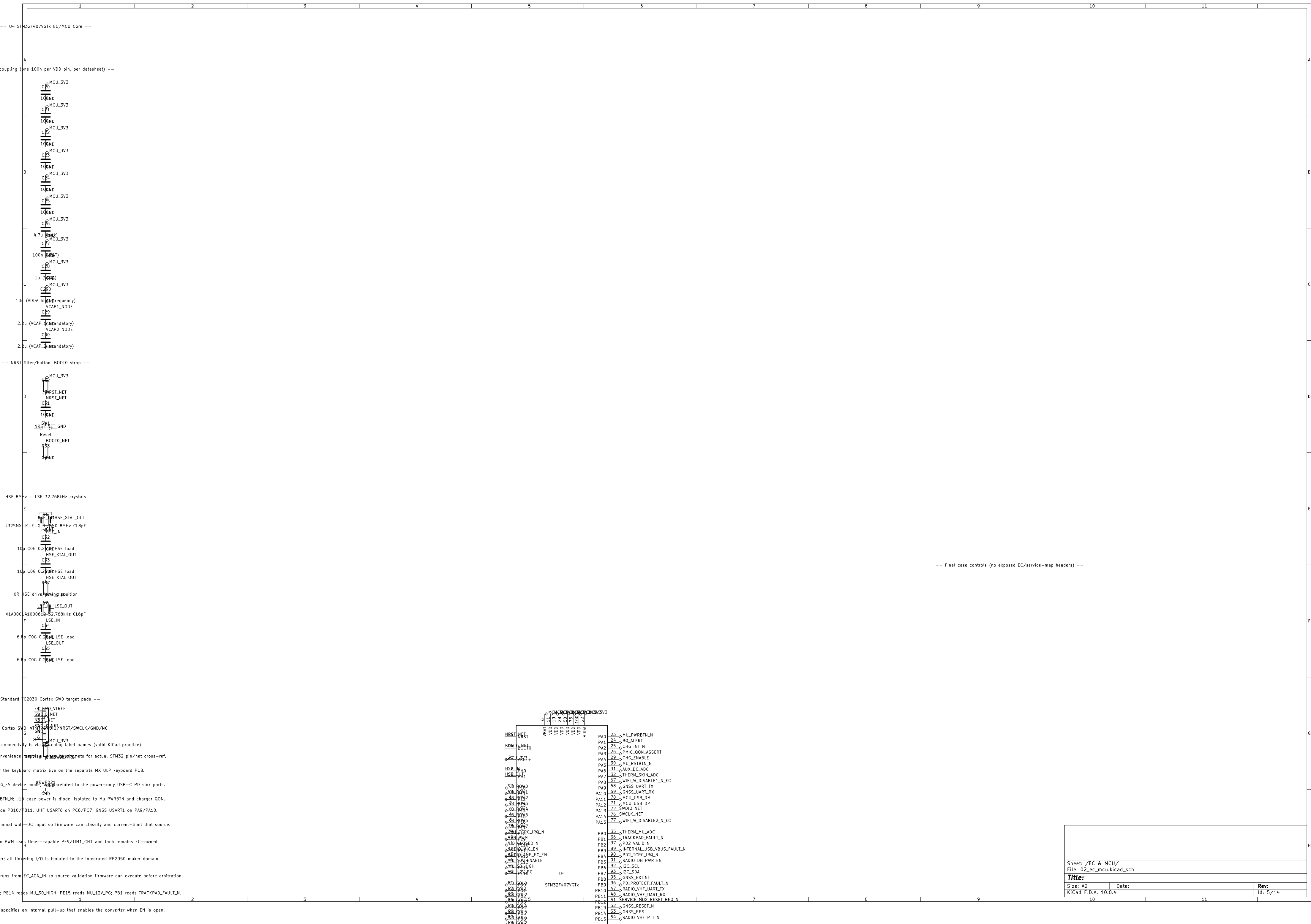
on SSD1306 modules: GND, VDD/3V3, SCK/SCL, SDA. Use 3.3V modules only.

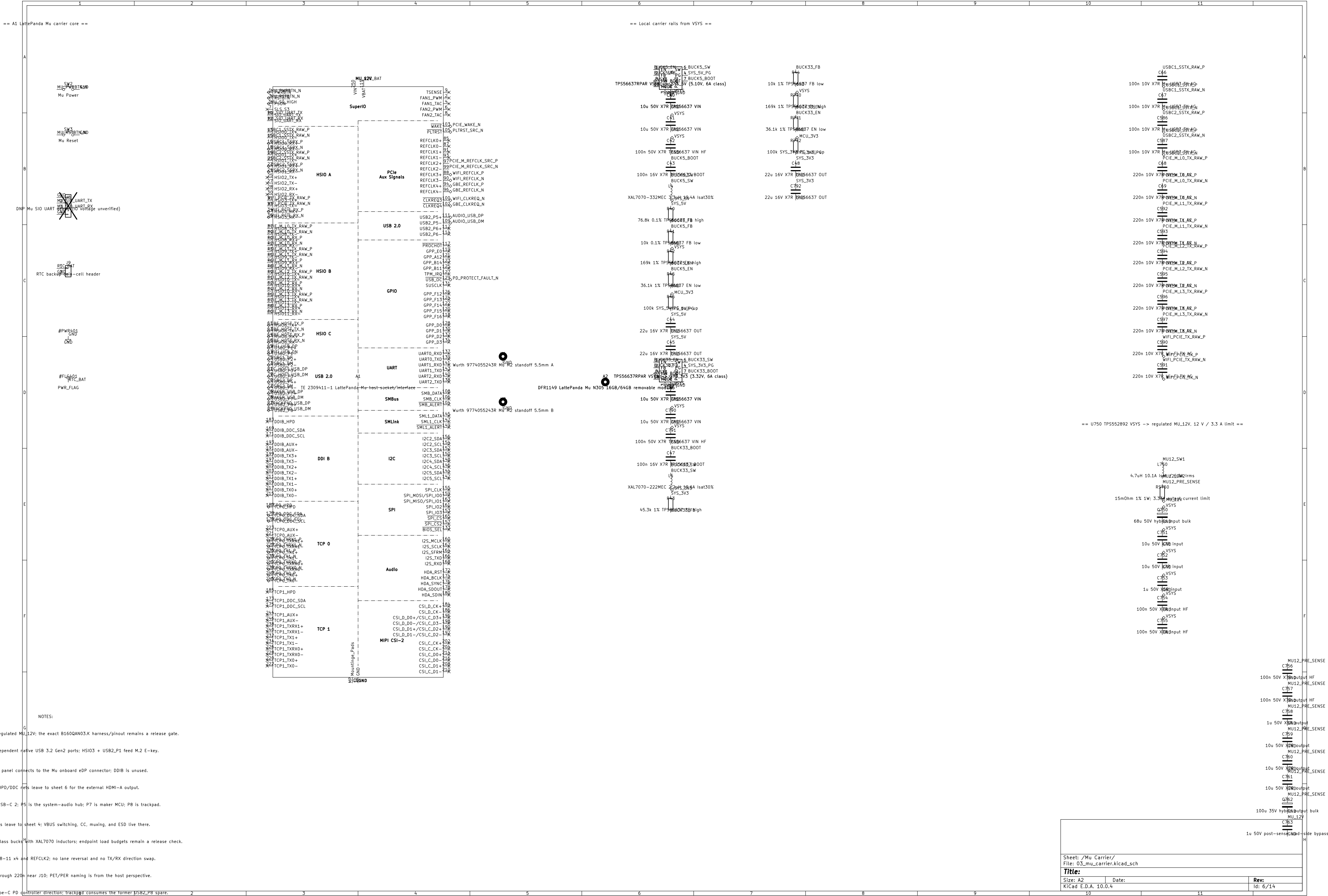
reset; firmware selects one TPS25751A channel and validates PD Status 0x35 plus PDO/RDO 0x31/0x32.

FI functions do not depend on radio-daughterboard presence or power.

L98/R199 keep WLAN and Bluetooth disabled until EC firmware explicitly enables them.

Sheet: /Wi-Fi/Bluetooth & OLEDs/ File: 07_radio_oled_gps.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 4/14





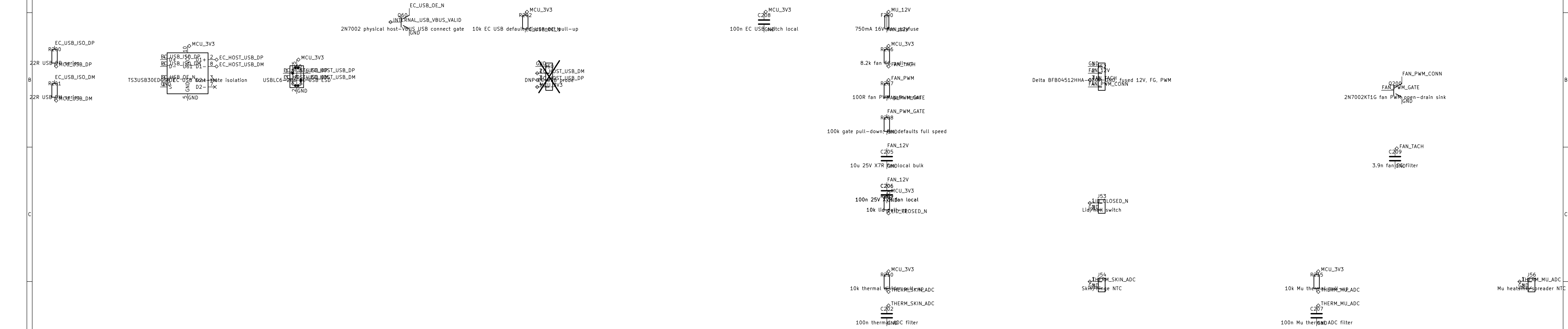
```
laptop services: EC USB, trackpad, fan, lid, thermal ==
```

re Mu USB2 ports and EC GPIO into internal laptop plumbing.

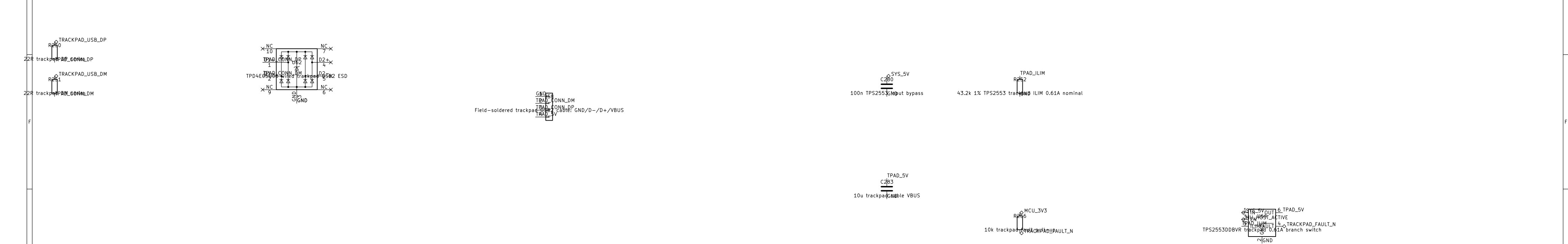
he Mu onboard eDP connector and needs no motherboard USB path.

```
EC USB device link to host Mu USB2_P3 ==
```

== Fan, lid, and thermal service ==



```
Required internal trackpad on Mu USB2_P8 ==
```



-, 3=D+, 4=VBUS. The USB-C plug's compliant 56k R_p handles trackpad-side CC; U64 provides current limiting and host-off isolation.

NOTES:

_VBUS is above the supervisor threshold. U61 defaults disconnected so an always-powered EC cannot back-drive an off host PHY.

panel is non-touch; USB2_P4 is paired with native USB-C port 2.

CKPAD_FAULT_N is pulled up to the EC; S3 removes trackpad VBUS, so wake-from-trackpad is not supported.

has a bench fixture/fallback and is not populated on the motherboard.

interface, and 25kHz open-drain PWM. Floating PWM commands full speed; firmware never drives the fan PWM node high.

ses/revolution, and 0% PWM stops the fan. Gate pull-down makes

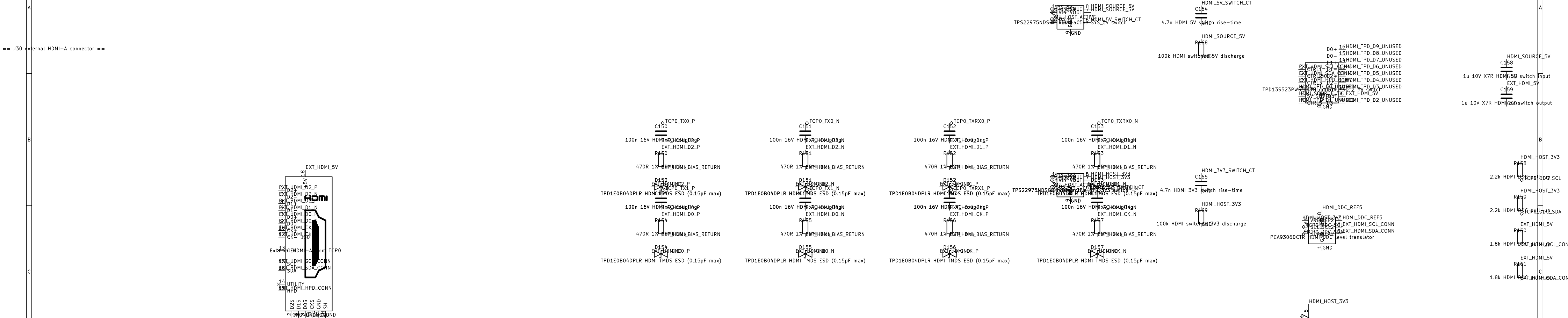
arate skin/hinge and Mu heatsink NfCs on ADC-capable EC pins.	2	3
---	---	---

Sheet: /Internal Services/ File: 08_internal_services.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 7/14

```
== TCP0 external HDMI 2.0 output ==
```

0 as HDMI 2.0; the internal panel uses the Mu module's onboard eDP connector.

gating, DDC/HPD translation, and 5V isolation follow the Mu reference.



NOTES:

retired Intel® controller is a bench-test/fallback fixture, not motherboard circuitry.

1.6V TPD1E0B04DPLR single-line parts; route each shunt at J30 with no stub.

used TPD13S523 legacy channels use the datasheet 75R-to-GND termination.

er while the Mu is off; PCA9306 translates DDC/SCDC and U53 buffers HPD.

Differential, Match pair skew to under 5 mil per Mu HDMI guide (current ~495 mil).

ntial. Match data-pair skew under 5 mil and refclock under 5 mil per Mu PCIe guide.

pin 18 under 55mA and validate the finished HDMI 2.0 layout on hardware.

nal radio/GNSS/audio daughterboard interface ==

boots and operates normally with J2300 completely unpopulated.

RADIO_DB_5V only after the passive PRESENT_N strap is detected.

Default-off protected daughterboard power ==

100k radio eFuse/enable pulldown

1.65k 0.1% radio eFuse/ILM about 2.0A

4.7n radio eFuse/controlled rise

10u SYS_5V radio eFuse input

100n SYS_5V radio eFuse input HF

22u RADIO_DB_5V output bulk

100n RADIO_DB_5V output HF

to phantom device or back-power when daughterboard is absent ==

100k USB switch/enable

2N7002K radio eFuse USB enable

100n USB switch local

Safe defaults for every off-board control ==

10k VHF RX default low

10k UHF RX default low

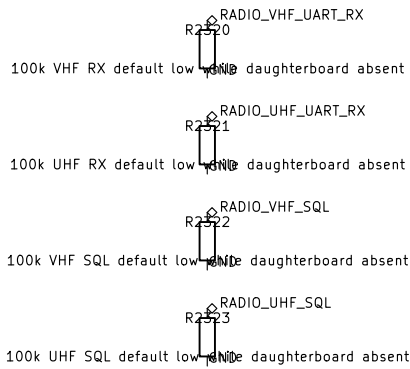
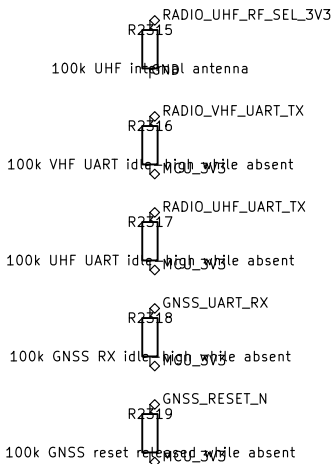
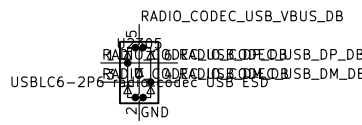
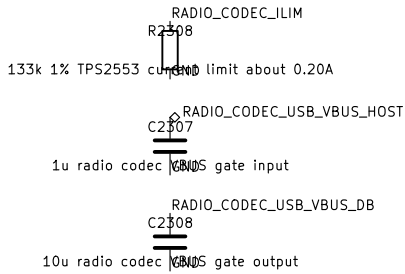
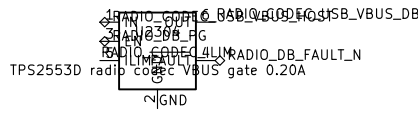
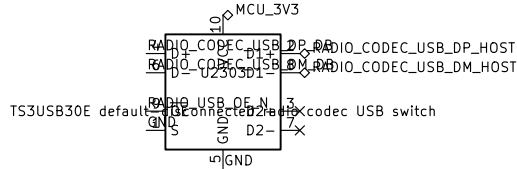
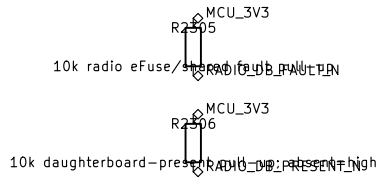
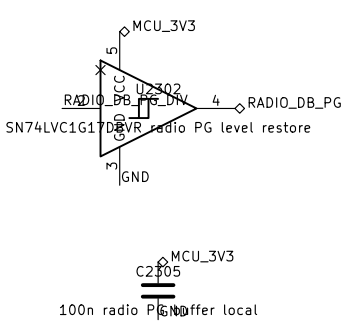
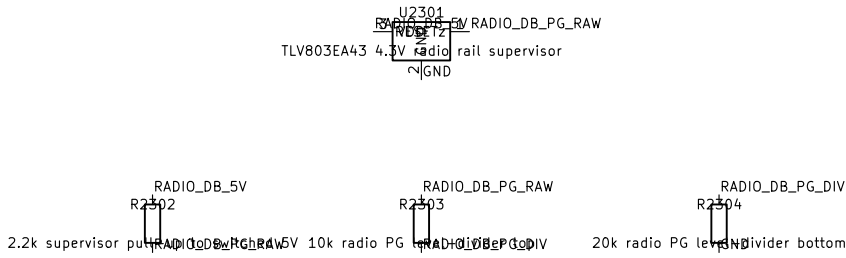
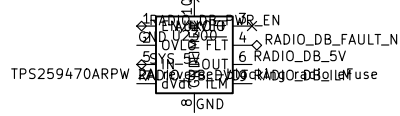
10k VHF SQL default low

10k UHF SQL default low

100k VHF internal antenna

ing series resistors at the removable-board boundary ==

use DF40 removable radio daughterboard connector ==

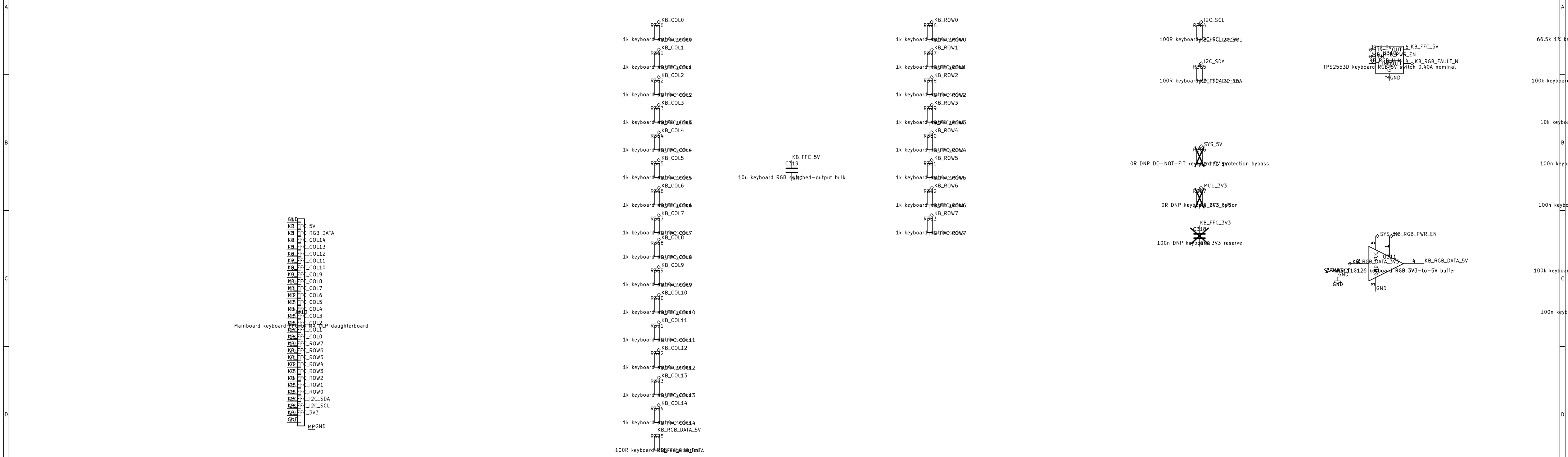


Sheet: /Optional Radio Daughterboard Interface/ File: 09_radio_daughterboard_interface.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 9/14

== Mainboard keyboard FFC Interface ==

connector only: the MX ULP switches and diodes live on the separate keyboard PCB.

kicad_pcb imports only the laptop-side FFC connector, not keyboard-switch footprints.



NOTES:

U320 for two top-side FH12 bottom-contact connectors and a Type-A same-side FFC.

0315, 30 circuits, 0.5 mm pitch, 30 mm, Type A / same-side contacts.

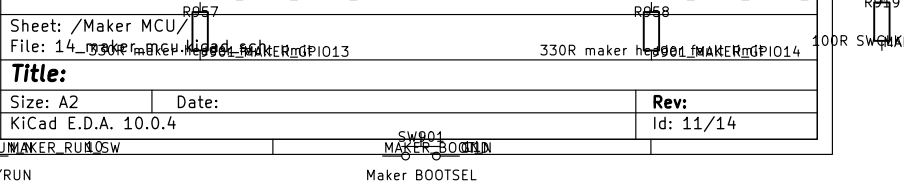
the future RGB rail below 0.5A/contact and defaults off; rev A has no RGB load.

repurposes unused COL15 for RGB data. Rows 5..7 and COL14 remain spare.

and firmware must enforce a 325mA LED budget; unrestricted full-white is unsupported.

et, SWD, ADC filtering, and decoupling follow Raspberry Pi Pico 2 guidance.

== Protected user-facing maker GPIO header ==



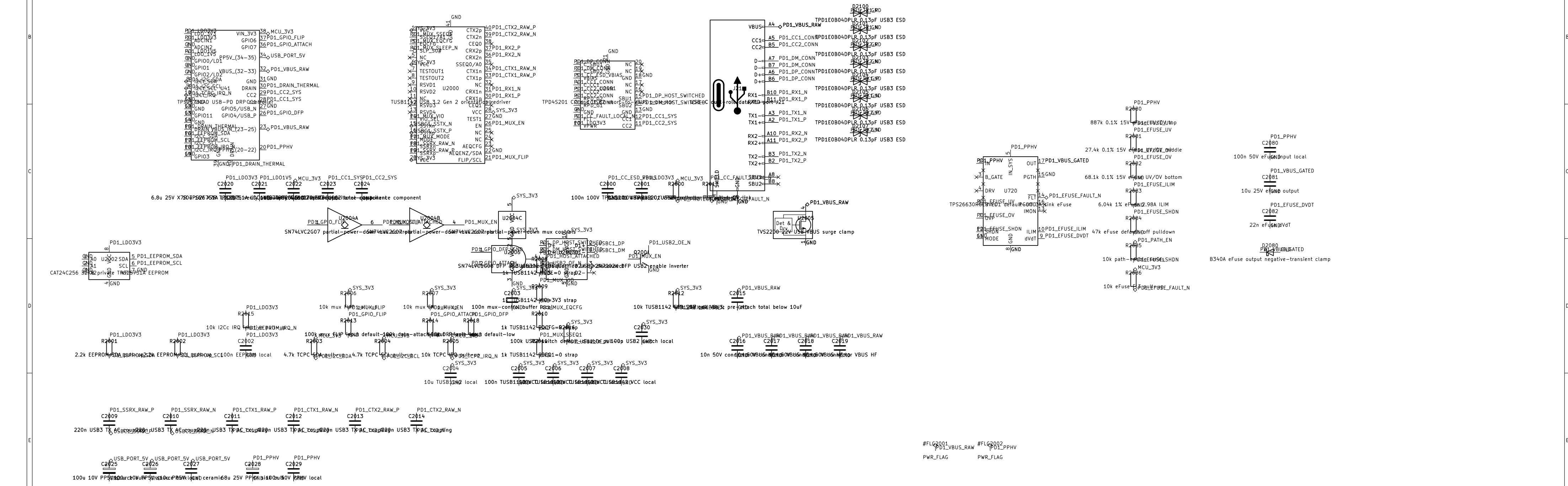
USB-C data/charging ports: J21 left and J11 right ==

USB2 and can sink negotiated 15V or source protected default USB current.

while powered from dead-battery VBUS; no EC is needed to obtain the first 15V contract.

- C pre-attach capacitance limit; large sink bulk is behind the integrated PD switch.

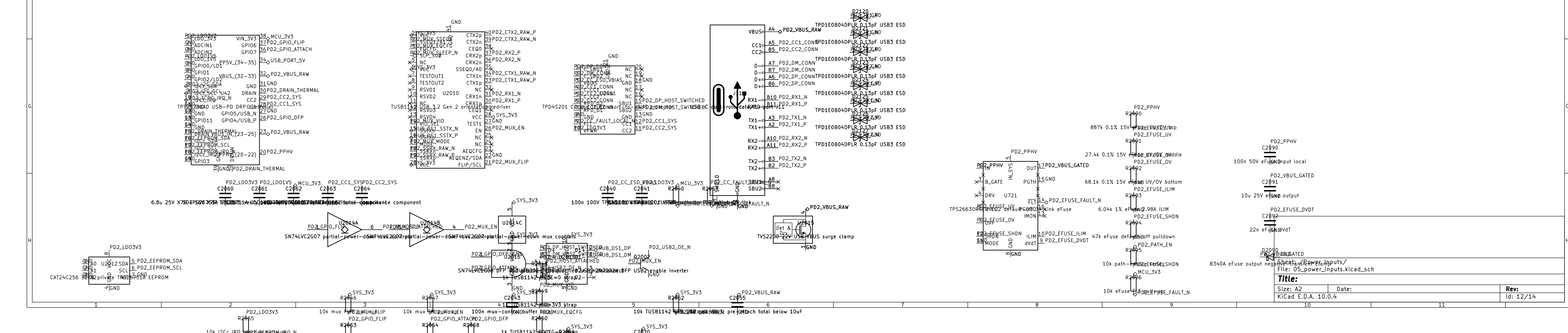
3.2 Gen 2 data plus USB-PD charging, side port 1 ==



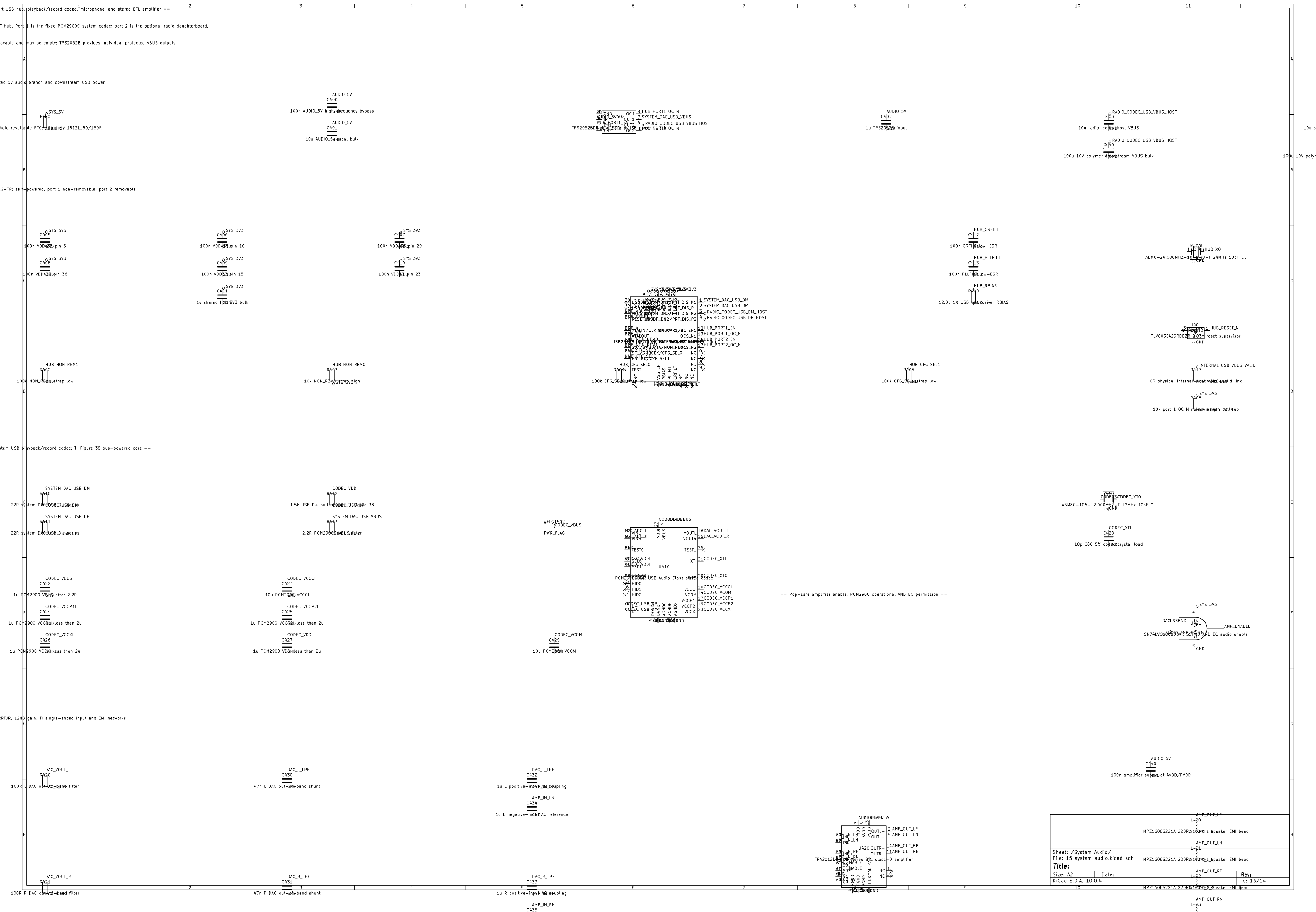
3A sink, 5V/0.9A source, default Rp, GPIO6 FLIP, GPIO4 DFP, and GPIO7 data-attached.

and SuperSpeed. They remain disconnected through reset, detach, and sink-only operation.

3.2 Gen 2: Gate plus USB-FD charging, side port 2 ==



Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 12/14



10GbE Gigabit Ethernet on LatticePanda-M0 default HSI06 ==

Panda's DFR1142 Mu carrier reference: HSI06, REFCLK4, CLKREQ4, and PLT_RST.

through-hole mid-mount RJ45 with integrated magnetics; no external transformer is required.

RTL8111H: C500/C501 are at the Mu transmitter pads, not by U500 ==

== Four 100-ohm MDI pairs, low-capacitance ESD, and integrated-magnetics RJ45 ==

1.3.3V / internal 1.0V rails and reset-side straps ==

2pF legs implement the qualified 8pF load including about 2pF stray ==

PCIe device, place C500–C503 beside U500 per the Mu guide and route over an uninterrupted reference plane.

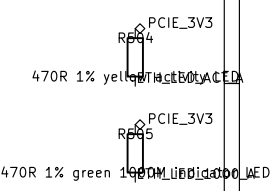
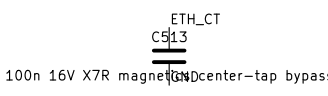
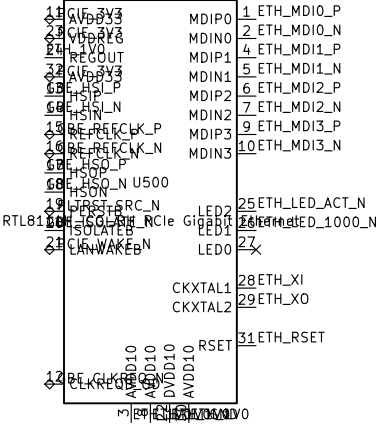
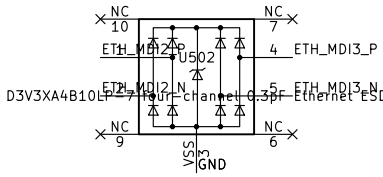
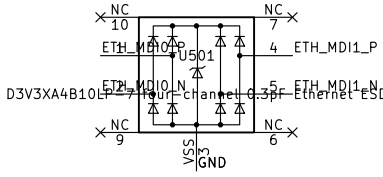
J502 immediately behind J500, avoid stubs, and do not route digital signals under the magnetics.

1.30mm; PCB/panel plane is 2.03mm behind the front face; panel opening 17.98 x 9.69mm.

common and confirm LED copper lands on one physical JXD1 sample before ordering the motherboard.

through-hole mid-mount RJ45 support and whether J500 is manual/post-reflow insertion.

M indication, not a generic link LED. Verify both jack LEDs at 10/100/1000 on hardware.



== Connector shield / aluminum chassis interface ==

